AI ACADEMY · LEVEL 7 · GRADE 12 · WEEK 3 · WORKBOOK

Finding primary research

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

One complete versioned source and claim trail.

Keep identity, artifact role, dataset/metric, locator, access status and unresolved mismatches together.

Keep one habit: Recall → W1–W3/W5 → independent source mismatch → literature-source project → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Reading a research claim. Give an example and a mistake that the idea helps you avoid.

Reading a research claim

> **My remembered idea, example and correction**
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> | |
> |---|
> | |
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MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Primary research reports a study’s own methods/results; secondary summaries help discovery but do not replace inspection. Record title, authors, version, date, identifier, access location and exact claim locator together with dataset/metric/denominator.

  Complete fictional main cards: BLOG-R newsletter v1 says 100 cases without supplied primary support; “Room Retrieval Pilot”, Fictional Team R, STUDY-R-v1 dated 2026-01-01 methods §2 says 40; revised supplement STUDY-R-v2 dated 2026-01-02 revised methods §2 says 60. These are internal classroom card labels, not real papers or identifiers.

  Keep version-specific counts; case membership/revision explanation and raw appendix are not supplied, so do not add or independently verify them. Separate BENCH-v1 fictional report methods §2 has ten fixed labelled tasks; Table 1 reports 9/10 correct. Blog “90% reliable” omits task-set scope. If original unavailable, retain secondary/unverified status and do not claim methods/results inspection.

  Paper/code/data/commentary are distinct artifacts; primary status itself does not ensure quality. No actual literature search or real citation is claimed.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 6

W2 · Trace the worked case

> Keep the supplied inputs, comparison rule and evidence together.

Complete task and evidence

- Primary research reports a study's own methods/results; secondary summaries help discovery but do not replace inspection. Record title, authors, version, date, identifier, access location and exact claim locator together with dataset/metric/denominator.

  Complete fictional main cards: BLOG-R newsletter v1 says 100 cases without supplied primary support; "Room Retrieval Pilot", Fictional Team R, STUDY-R-v1 dated 2026-01-01 methods §2 says 40; revised supplement STUDY-R-v2 dated 2026-01-02 revised methods §2 says 60. These are internal classroom card labels, not real papers or identifiers.

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  Paper/code/data/commentary are distinct artifacts; primary status itself does not ensure quality. No actual literature search or real citation is claimed.

- A blog says experiment had 100 cases; original methods say 40 and supplementary version says 60 after revision. Show the intermediate steps and give the conclusion with its scope.

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Primary research reports a study's own methods/results; secondary summaries help discovery but do not replace inspection. Record title, authors, version, date, identifier, access location and exact claim locator together with dataset/metric/denominator.

  Complete fictional main cards: BLOG-R newsletter v1 says 100 cases without supplied primary support; "Room Retrieval Pilot", Fictional Team R, STUDY-R-v1 dated 2026-01-01 methods §2 says 40; revised supplement STUDY-R-v2 dated 2026-01-02 revised methods §2 says 60. These are internal classroom card labels, not real papers or identifiers.

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  Paper/code/data/commentary are distinct artifacts; primary status itself does not ensure quality. No actual literature search or real citation is claimed.

- A report omits this qualification: Being primary does not guarantee soundness; methods and **evidence** still require assessment. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Primary research reports a study's own methods/results; secondary summaries help discovery but do not replace inspection. Record title, authors, version, date, identifier, access location and exact claim locator together with dataset/metric/denominator.

  Complete fictional main cards: BLOG-R newsletter v1 says 100 cases without supplied primary support; "Room Retrieval Pilot", Fictional Team R, STUDY-R-v1 dated 2026-01-01 methods §2 says 40; revised supplement STUDY-R-v2 dated 2026-01-02 revised methods §2 says 60. These are internal classroom card labels, not real papers or identifiers.

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  Paper/code/data/commentary are distinct artifacts; primary status itself does not ensure quality. No actual literature search or real citation is claimed.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A search result describes a benchmark score, but the linked paper uses a different dataset. What should the comparison record contain?

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the research evidence

> Preserve the question, protocol, observations, analysis and limits.

Complete task and evidence

- Add a labelled finding primary research **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of finding primary research.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	

Evidence	My record
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Finding primary research. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Finding primary research. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A search result describes a benchmark score, but the linked paper uses a different dataset. What should the comparison record contain?

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

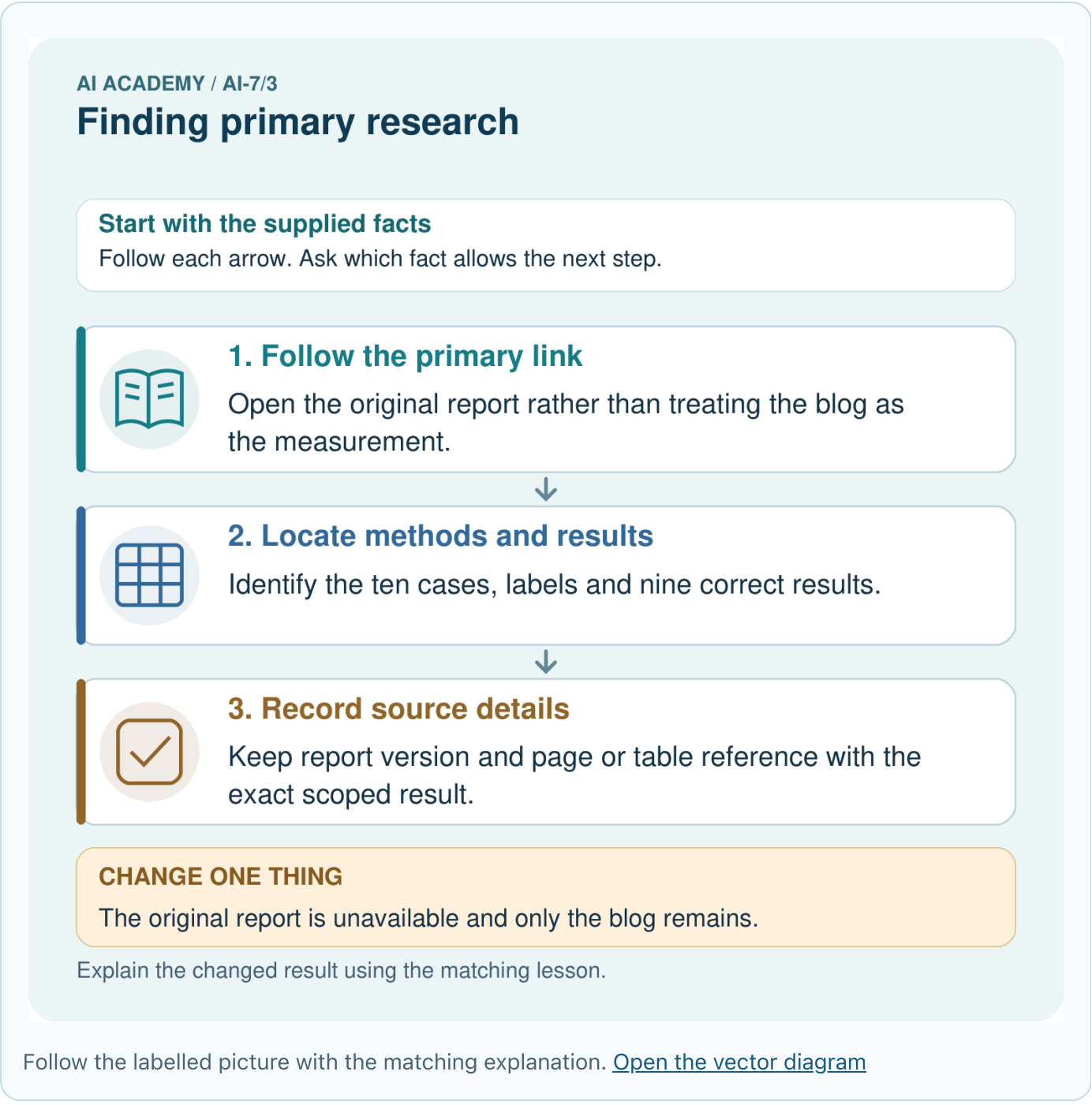
- Find the source behind a summary. A blog describes a fictional benchmark improvement and links an original study report.

- Blog says "90% reliable". Original report records 9/10 correct on one fixed fictional task set and describes its labels.

- Separate what the blog claims from what the original report records. Which methods and denominator must be inspected, and what status is honest if the original report is unavailable?

- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

  Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project milestone**

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Reproducible AI Research Study
- Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.
- For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Find the source behind a summary. A blog describes a fictional benchmark improvement and links an original study report.
- Which labelled step matches this detail? Keep report version and page or table reference with the exact scoped result.
- 1. Record source details
- 2. Locate methods and results

- 3. Follow the primary link

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Record source details Keep report version and page or table reference with the exact scoped result.

Primary research exposes the procedure and **observation**s behind a summary. It still needs scrutiny; being original does not make every interpretation correct.

Changed-case question and optional hint

- Changed-case practice: The original report is unavailable and only the blog remains. Which conclusion is supported by the supplied facts?

- 1. This one example guarantees correct results for every future **input**.

- 2. We can decide the result without examining the changed **input** or the procedure.

- 3. Mark the measurement as unverified rather than filling in absent methods.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Mark the measurement as unverified rather than filling in absent methods. The summary cannot supply details that have not been inspected.

Repair and retain the record

Review Reading a research claim (ai-7/2). Goal: Explain the **mechanism** and terminology of reading a research claim. Short practice: Calculate the baseline and new rates for 2/10 and 4/10.

Explain both twice the rate and the percentage-point change, and say why neither measures every learner's improvement.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

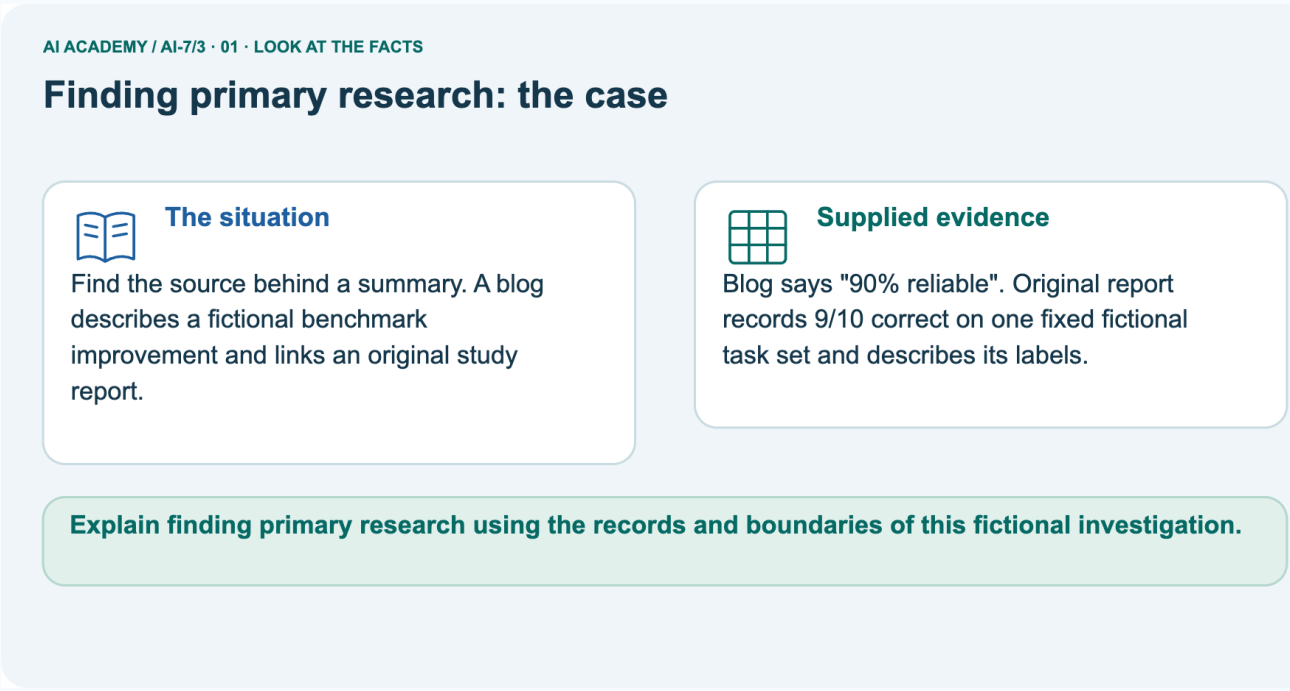
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Finding primary research: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

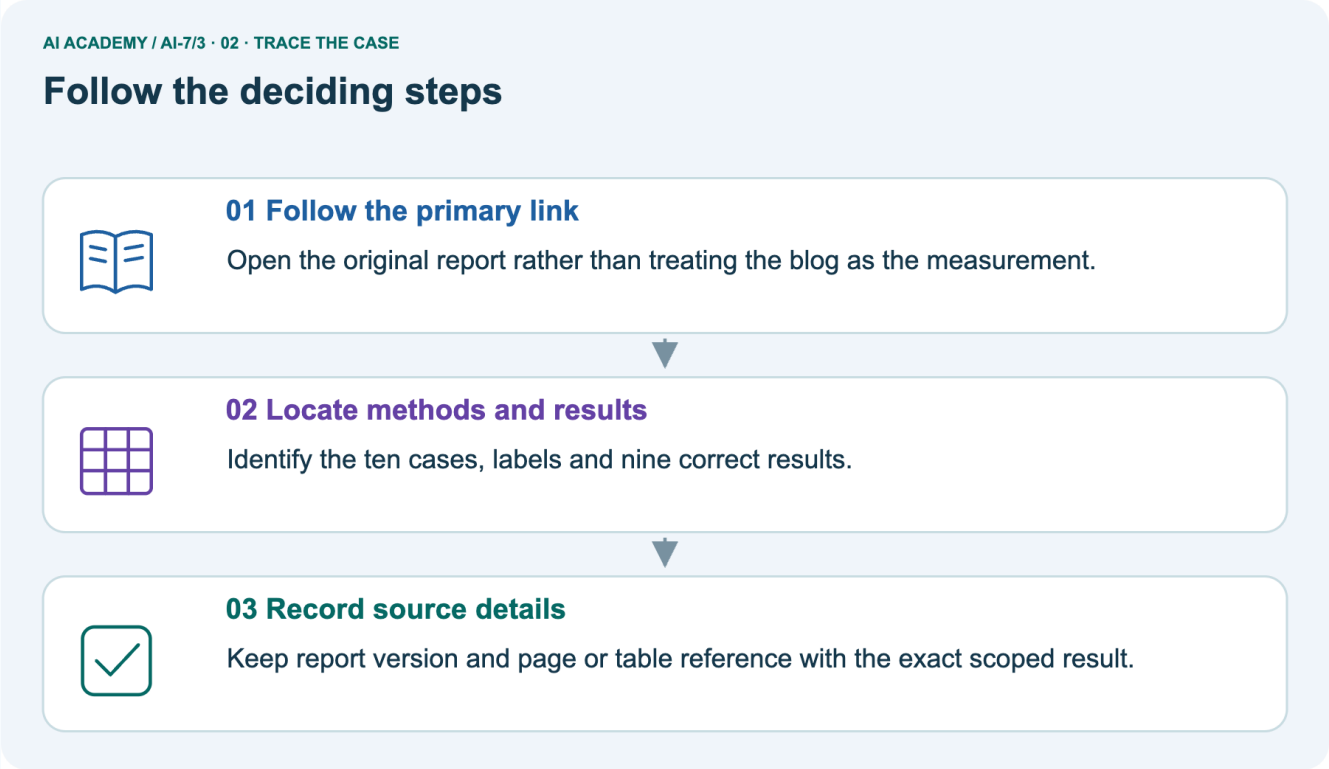
Find the source behind a summary. A blog describes a fictional benchmark improvement and links an original study report. Supplied facts: Blog says "90% reliable".

Original report records 9/10 correct on one fixed fictional task set and describes its labels.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



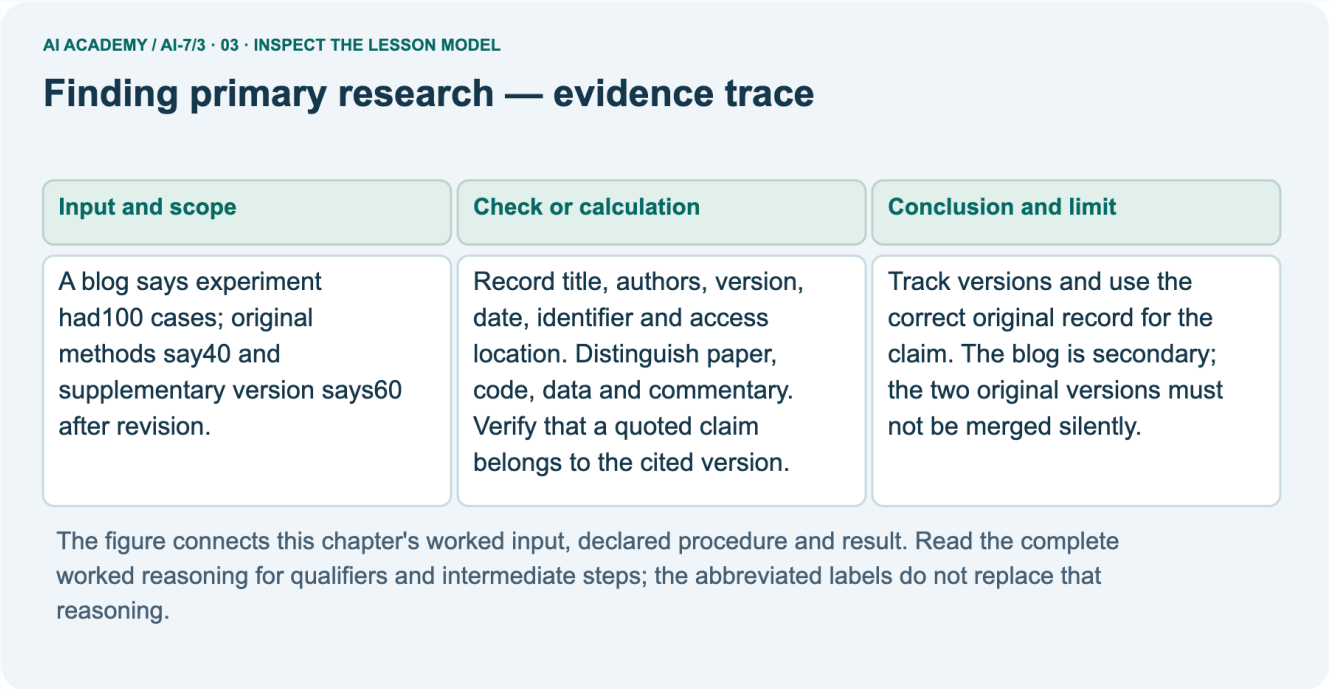
Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the primary link: Open the original report rather than treating the blog as the measurement. → Locate methods and results: Identify the ten cases, labels and nine correct results. → Record source details: Keep report version and page or table reference with the exact scoped result.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Finding primary research — evidence trace



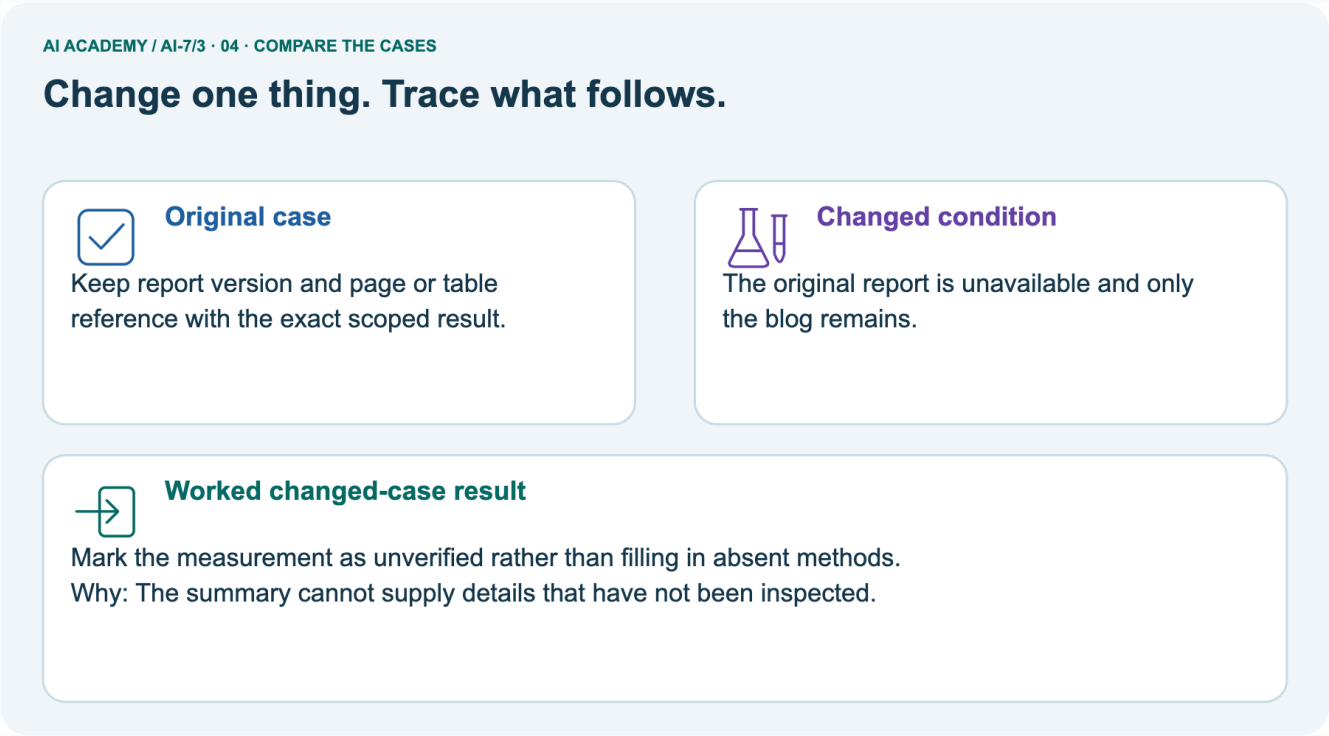
Follow the labelled picture with the matching explanation. Open the vector diagram

Finding primary research — **evidence** trace. A blog says experiment had 100 cases; original methods say 40 and supplementary version says 60 after revision. | Record title, authors, version, date, identifier and access location. Distinguish paper, code, data and commentary. Verify that a quoted claim belongs to the cited version. | Track versions and use the correct original record for the claim. The blog is secondary; the two original versions must not be merged silently.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

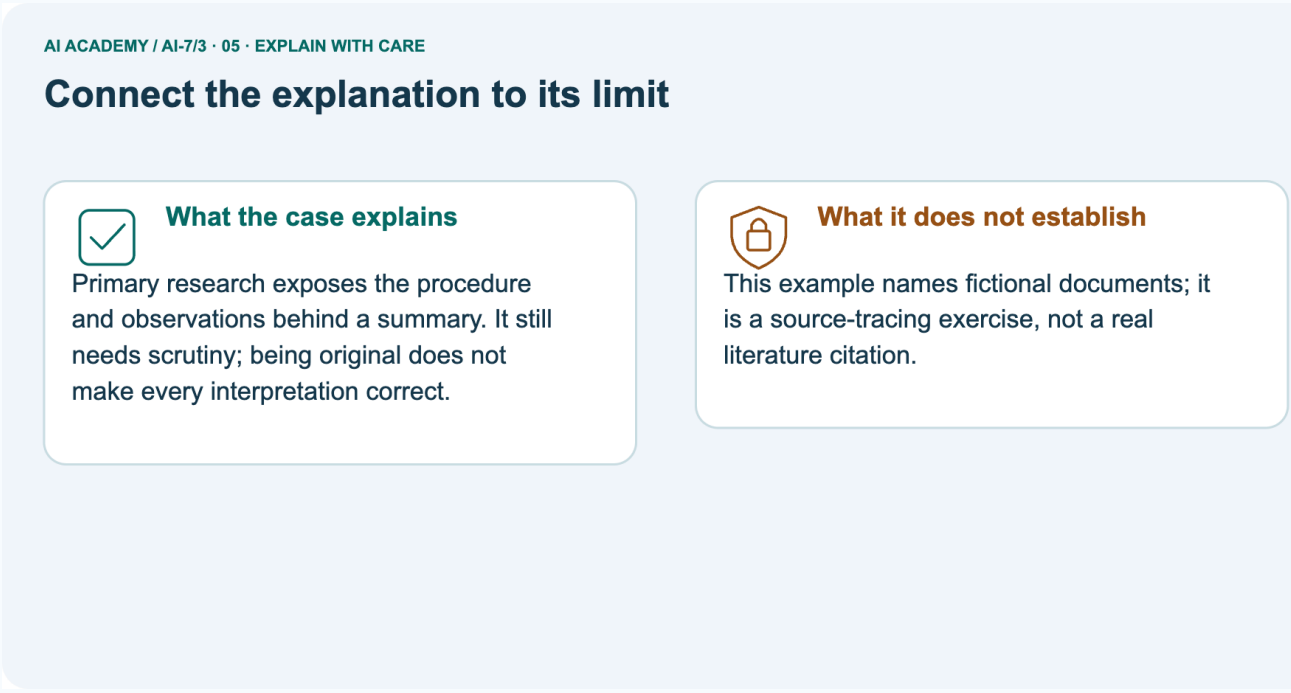
Original case: Keep report version and page or table reference with the exact scoped result.
Changed condition: The original report is unavailable and only the blog remains.

Expected result: Mark the measurement as unverified rather than filling in absent methods.
Reason: The summary cannot supply details that have not been inspected.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

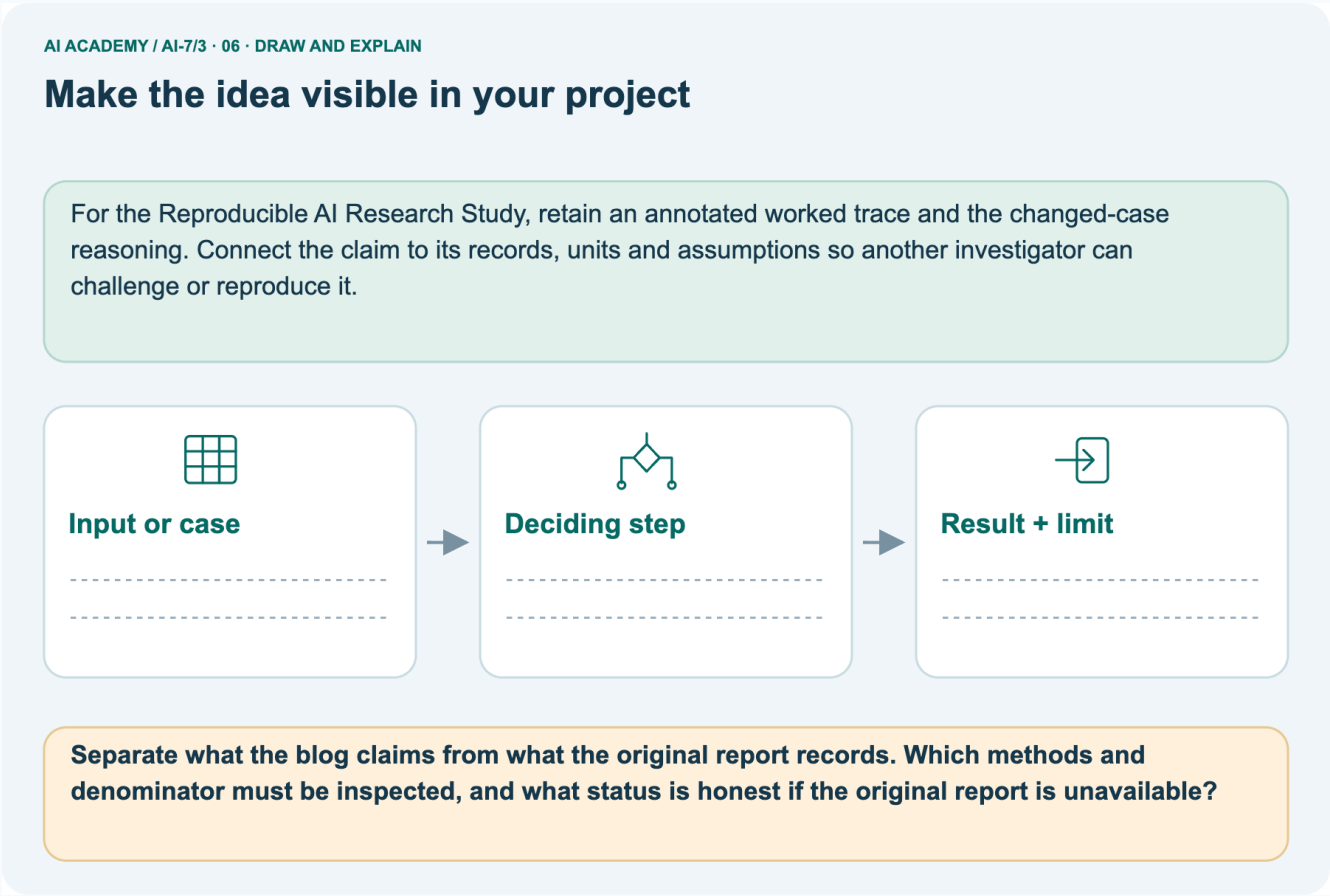
Primary research exposes the procedure and **observation**s behind a summary. It still needs scrutiny; being original does not make every interpretation correct.

Limit: This example names fictional documents; it is a source-tracing exercise, not a real literature citation.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Separate what the blog claims from what the original report records.

Which methods and denominator must be inspected, and what status is honest if the original report is unavailable?

Draw your project trace and justify the deciding step using **evidence**.

> **My picture explanation, deciding evidence and one limit**

▼ **Earlier wording, alternatives and picture transcripts**

01. Finding primary research: the case

02. Follow the deciding steps

03. Finding primary research — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ **Optional interactive practice and completed work**

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.